Recent Advances in Responsible AI
Introduction to Differential Privacy

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What Do We Mean by Privacy?

Universal Declaration of Human Rights (Article 12)

No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation.

Legal Foundations of Privacy General Data Protection Regulation (GDPR, 2016/679)

- Principles: Lawfulness, fairness, transparency, data minimization, security.
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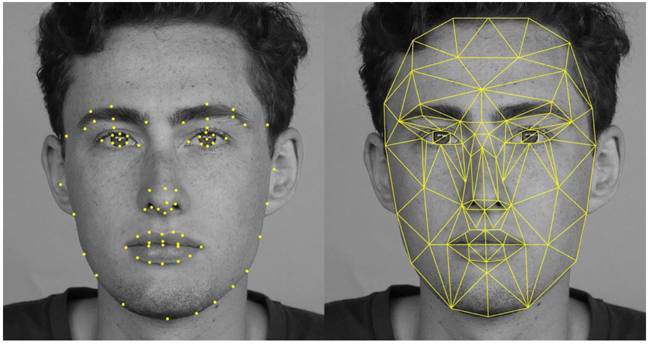
These laws cover the collection, use and disclosure of PI, but they may not capture the spirit of what privacy should mean in data analysis.

Where should we care? (Personal Data & Surveillance)

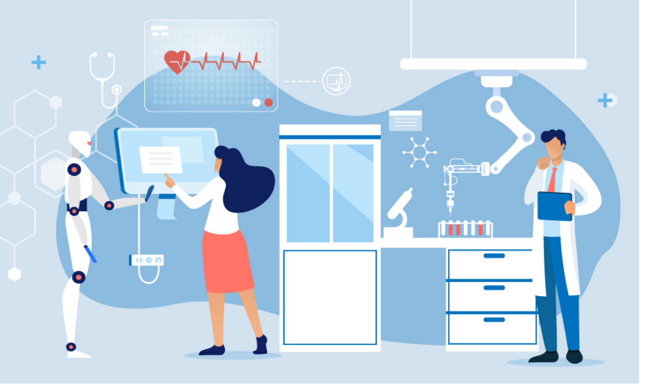
Facial Recognition in Public Spaces

- AI identifies individuals in real time.
- Used for security, but raises concerns about surveillance.



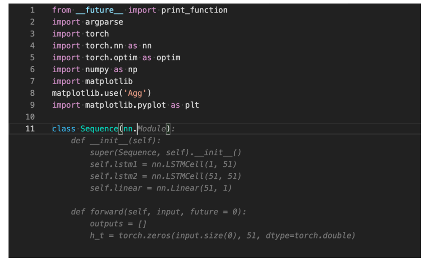
Healthcare and Personal Data

- A user shares sensitive medical information.
- How is this data stored and used?

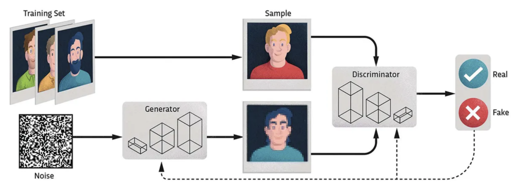


Is This Private? (AI-Generated Content & Code Privacy)

GitHub Copilot and Code Privacy Copilot Autocomplete memorizes and repeats blocks of code from repos. Could it expose private code?



AI-Generated Text and Data Privacy GenAI learns to generate content that is similar to a training set. Has the privacy of training set samples been breached?



What privacy is not ?

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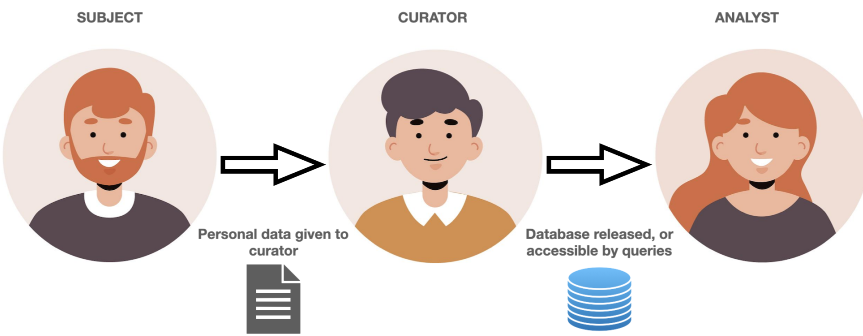
- Cryptography is a **security protocol**. Our message is secure from prying eyes. But the entire content of our message is revealed to the recipient.
- Cryptography is **binary** - either all information is revealed, or none is

Setting for Defining Privacy

- **Subjects** - individuals who give, or refuse to give, their personal data
- **Curator** - a trusted centralized party that holds data and controls the database
- **Analysts/Adversaries** - access the database for good or for evil



Privacy is a promise from the curator to the subjects that **no harm** will come to them by including their data in the database.

The Promise of Privacy – Teaching vs Participation

Curator's promise [Dwork, 2014] - *You will not be affected, adversely or otherwise, by allowing your data to be used in the database, no matter what other data sets are available.*

Imagine an individual choosing to provide their data to a study on smoking.

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How can we learn something about a population, while not affecting individuals?

Differential Privacy

Informal Definition

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- Thus looking at the output can't tell if a user was in the dataset or not.
- If you can't even know if a user is present, you can't know their data

Attempts at Privacy - Anonymisation Fiascos

Some approaches are commonly used, but don't preserve the promise.

The curator may remove personal information, and publish the rest.

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Disturbing Headlines and Paper Titles

- *A Face Is Exposed for AOL Searcher No. 4417749* [Barbaro & Zeller, 2006]
- *Robust De-anonymization of Large Datasets (How to Break Anonymity of the Netflix Prize Dataset)* [Narayanan & Shmatikov, 2008]
- *Matching Known Patients to Health Records in Washington State Data*[Sweeney, 2013]
- Harvard Professor Re-Identifies Anonymous Volunteers In DNA Study [Sweeney et al. 2013]

Attempts at Privacy - Anonymisation

Linkage Attacks

It can be possible to de-anonymize records using outside information.

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Netflix released data on sparse user movie ratings - de-anonymized by comparing to public ratings on IMDB. [Narayanan & Shmatikov 2008]

Name	Age	Job	Salary
[redacted]	23	Clerk	50,000
[redacted]	45	Driver	60,000
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Why is data Anonymisation hard?

- High-dimensional/high-resolution data is essentially unique:

office	*department*	*date joined*	*salary*	*d.o.b.*	*nationality*	*gender*
London	IT	Apr 2015	£####	May 1985	Portuguese	Female

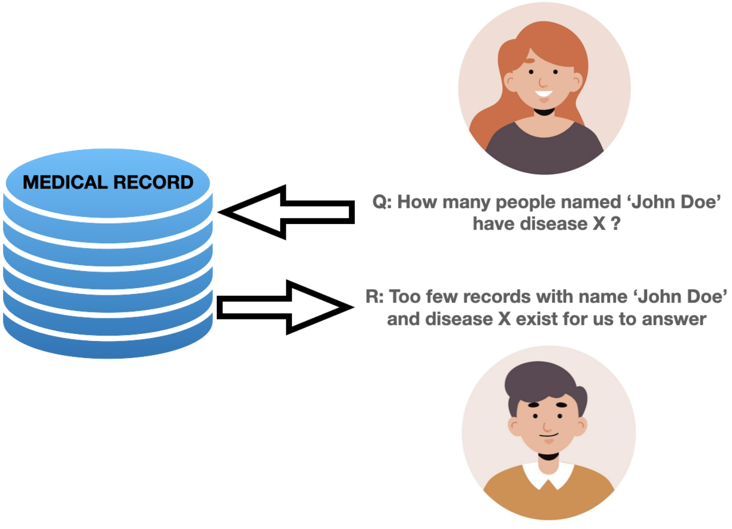
- Lower dimension and lower resolution is more private, but less useful:

office	*department*	*date joined*	*salary*	*d.o.b.*	*nationality*	*gender*
UK	IT	2015	£####	1980-1985	—	Female

Attempts at Privacy - Restrict Queries

The curator does not release the database, but will answer certain queries.

To avoid harm to individuals, queries will only be answered if they involve a large enough group



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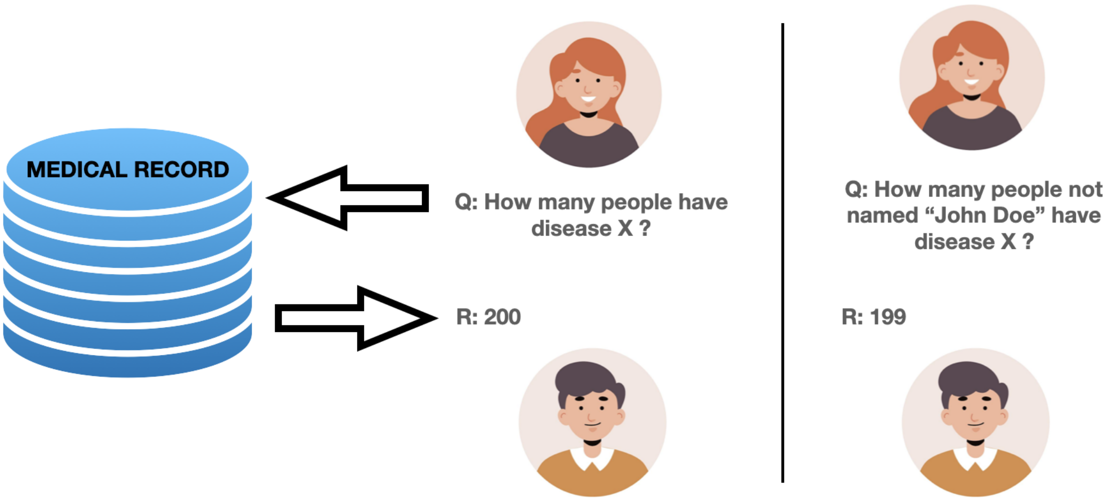
Differencing Attacks

If an individual is known to be in the database then 2 queries over large sets will reveal info about the individual.

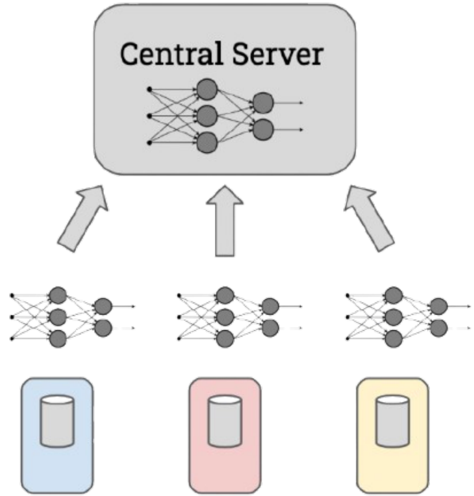


Attempts at Privacy - Federated Learning

Federated Learning is a distributed ML approach where data is not stored on a centralized server.

Instead, the model is trained on the device where data is collected (e.g. smartphone), and updates are aggregated by the central server.

Intuitively this seems more private, since the data is kept on the personal device where it is generated.



Clients with local, private data send model updates only.

Attempts at Privacy - Federated Learning

Memorization

Except, there are many examples of neural networks memorizing individual training examples which can be recovered by analyzing the model.
[Fredrikson, Jha, Ristenpart 2015]

Privacy as a resource

Fundamental Law of Recovery

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But if we cannot learn anything about anyone, then there is no information to be gained at all!

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But if we cannot learn anything about anyone, then there is no information to be gained at all!

An example of general information we would like to extract is: "20% of the data population are diabetic".

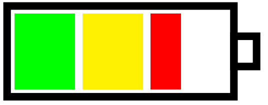
But now we have learned information about individuals - each individual belongs to a population where the prevalence of diabetes is 20%.

Name	Diabetic
Joe	No
Jun	No
Jaya	No
Janet	No
Jaro	Yes

Privacy Budget: A Limited Resource

Privacy is not binary (private or not). Instead, it is gradually consumed when we release information.

- Think of privacy like a **budget**: every time you share data, you "spend" some privacy.
- The more queries we make on private data, the more we reveal.
- Differential Privacy controls this by limiting the "cost" of each query.



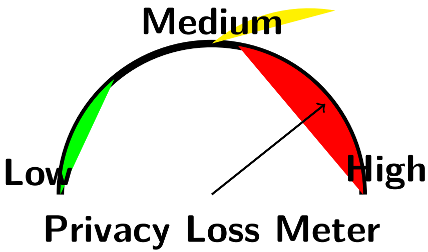
Privacy Budget

How Do We Spend Privacy?

The more accurate or detailed the answer, the more privacy is consumed.

Examples:

- **Low-cost queries** (small privacy loss)
  - "What is the average salary?"
  - "How many people in this dataset are over 30?"
- **High-cost queries** (large privacy loss)
  - "What is Alice's salary?"
  - Repeatedly querying slightly different groups to deduce individual info.



Differential privacy gives a formal definition of this resource.

Randomized Response: Adding Noise to Preserve Privacy

Analogy: Imagine asking people a sensitive question:

Have you committed tax fraud?

$\rightarrow$ Nobody would truthfully respond to this survey and incriminate themselves.

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Answer
Truthfully

Flip Again
If heads: Yes
If Tails: No

- The overall statistics remain useful: analyst gets an estimate of the true answer by understanding random process
- Individual responses are protected.

Randomized Queries for Privacy

Key Idea Instead of always returning the exact result, we introduce randomness.

Example: Counting Queries

- **Without randomness:** "How many people in the dataset earn >50k?"
- **With randomness:** Add small noise to the count.

Exact Count: 125 $\longrightarrow$ Noisy Count: 127

Differential Privacy works with randomized queries that return answers from a database.

Instead of returning the true answer to a query, each possible answer (including the true one) has some probability of being returned.

Differential Privacy (Formal Definition)

Consider two databases $\mathcal{D}$ and $\mathcal{D}'$ which differ by one record, and a randomized query $\mathcal{M}$ which acts on databases to give a result.

The query is **differentially private** if the results are **almost indistinguishable** for all databases that differ by one record.

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A query is (ϵ, δ)-differentially private if for all subsets of the output $\mathcal{S} \subseteq Range(\mathcal{M})$ and all pairs of databases that differ by one record

$$\mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{S}) \leq \exp(\epsilon)\mathbb{P}(\mathcal{M}(\mathcal{D}') \in \mathcal{S}) + \delta$$

Intuitively, the likelihood of any result from the query must be almost unchanged when one datapoint is added or removed.

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Counterexample A query that returns the exact count of records that satisfy property P. On neighbouring databases

$$\mathbb{P}(\mathcal{M}(\mathcal{D}) = n) = 1, \qquad \mathbb{P}(\mathcal{M}(\mathcal{D}') = n) = 0$$

so this exact count query cannot be DP unless $\epsilon = \infty$ and $\delta = 1$

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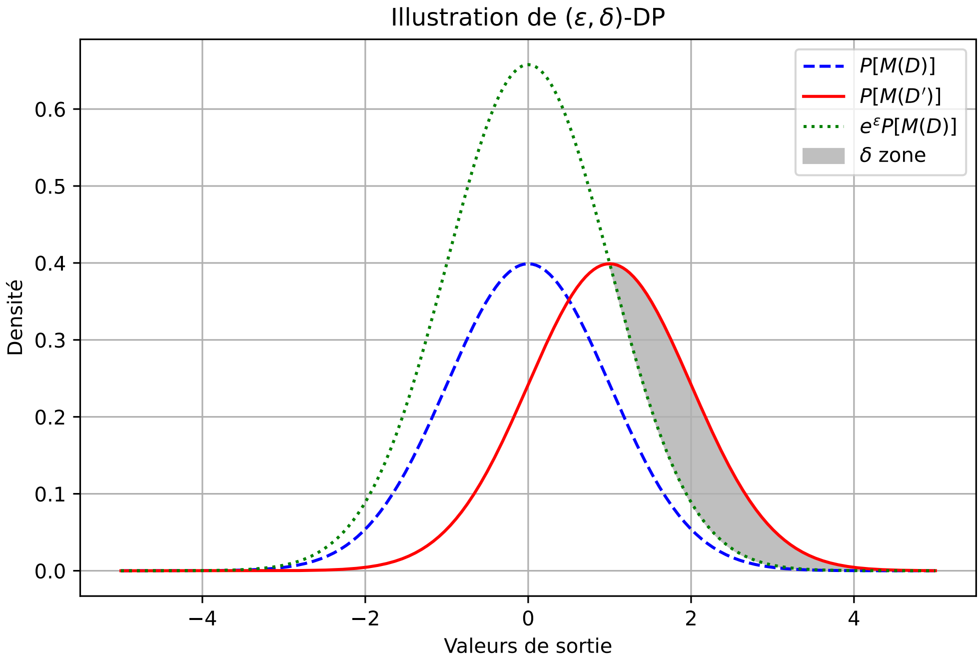
- Randomized - Any DP-query must be randomized
- Quantitative - ϵ and δ are numerical, where lower means better privacy
- Worst Case - We assume nothing about what the input datasets are like

Illustration of the bound



The Laplace Mechanism

Key Idea Add noise from the Laplace distribution to a numeric query result.

$$\mathcal{M}(D) = f(D) + \mathsf{Lap}\left(\frac{\Delta f}{\epsilon}\right)$$

where $f(D)$ is the exact query result and Δf is the *sensitivity* of f, i.e. the greatest possible variation in f when a single individual in the database changes

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Example – Counting the number of people over 30 years old

- Let $f(D)$ be the number of people older than 30 in a database.
- If one person is added or removed, $f(D)$ changes by at most 1. So, $\Delta f = 1$.
- We add noise $\eta \sim \text{Lap}(1/\epsilon)$ to ensure privacy.

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Example 2– Computing the average salary

- Let $f(D)$ be the average salary in a database.
- If salaries are in $[0, 100k]$, adding or removing one person can change the mean by at most $100k/n$.
- So, $\Delta f = \frac{100k}{n}$, meaning noise is **inversely proportional to database size**.

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Privacy vs. Accuracy Tradeoff

- large ϵ (e.g., 5) $\rightarrow$ less noise, but weaker privacy.
- small ϵ (e.g., 0.1)$\rightarrow$ more noise, but stronger privacy.
- In practice, $\epsilon \in [0.1, 1]$ is often used for strong privacy.

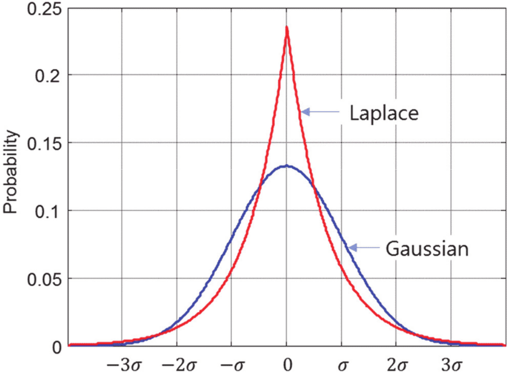
The Gaussian Mechanism

Definition: The Gaussian mechanism adds noise $\mathcal{N}(0, \sigma^2)$ to the query result.

Mechanism:

$$\mathcal{M}(D) = f(D) + \mathcal{N}(0, \sigma^2)$$

The standard deviation σ is chosen to satisfy (ϵ, δ)-DP.



Intuition behind δ in Differential Privacy

The Gaussian mechanism cannot guarantee a true $\epsilon - DP$ because the tail of the Gaussian distribution does not decay as fast as a Laplace distribution.

What does δ mean?

- Ensures that DP holds with probability at least $1 - \delta$.
- Small probability δ where privacy guarantees might fail.
- Necessary for Gaussian noise mechanisms.

How to choose δ? δ should be very small, ideally much smaller than the inverse of the dataset size:

$$\delta \ll \frac{1}{|D|}$$

A common heuristic is to set δ as:

$$\delta = O\left(\frac{1}{|D|^{1+c}}\right), \quad c > 0$$

Differentially Private SGD (DP-SGD)

Goal: Train machine learning models while ensuring differential privacy.

- Standard SGD updates model parameters based on gradients computed from mini-batches.
- DP-SGD introduces modifications to ensure privacy.

Key Modifications:

- Gradient Clipping
- Adding Noise through the Gaussian mechanism.
- Tracking Privacy Loss

Remark: the Gaussian mechanism is preferred because it adapts better to multidimensional gradients and offers greater stability.

Gradient Clipping

Why Clip Gradients?

- Prevents large gradients from leaking too much information about individual data points.
- Limits the sensitivity of each update, making it easier to add controlled noise.

How? Each gradient is clipped to a fixed norm C:

$$g \leftarrow g \cdot \min\left(1, \frac{C}{\|g\|}\right)$$

Intuition:

- Ensures that no single data point has an outsized influence on the training.
- Acts as a form of regularization, helping model stability.

Adding Noise to Gradients

Why Add Noise?

- Hides contributions of individual data points.
- Provides formal privacy guarantees under DP.

Mechanism: After clipping, Gaussian noise is added:

$$\tilde{g} = g + \mathcal{N}(0, \sigma^2 C^2)$$

where σ is calibrated based on the desired privacy level.

Tracking the DP Loss

- DP affects convergence due to **clipping** and **noise addition**.
- The training loss evolves differently compared to standard training.
- Important for diagnosing learning behavior under privacy constraints.

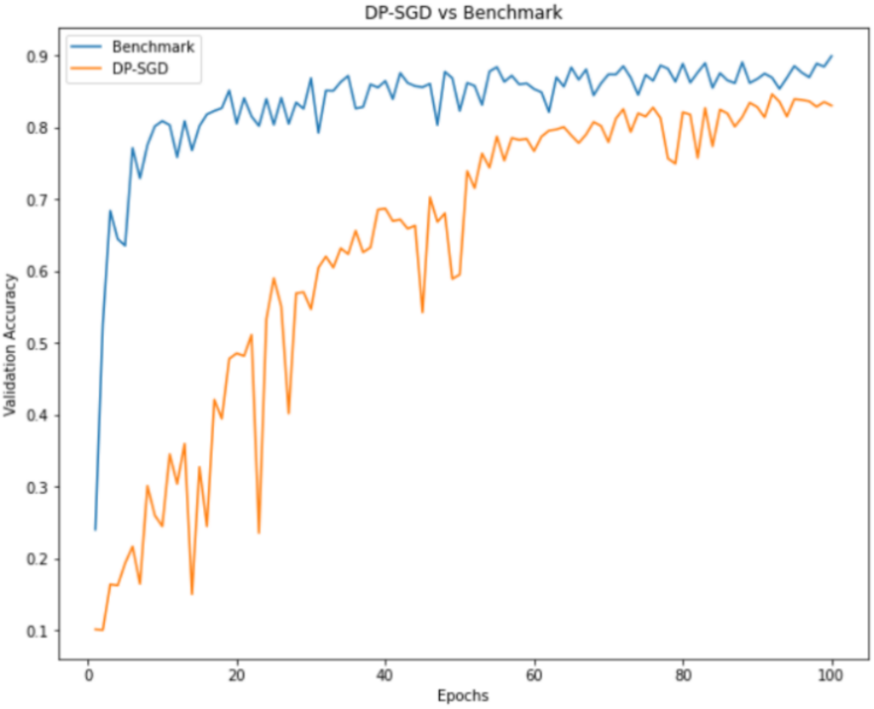
Key Observations

- DP loss decreases more slowly due to noisy updates.
- Clipping may limit the model's ability to learn fine-grained patterns.
- A balance is needed between privacy strength and model utility.

Comparison

- Non-DP training: loss decreases smoothly.
- DP training: loss fluctuates due to noise, converges more slowly.

SGD versus DP-SGD



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  Example: Even if an attacker knows a user's age and gender, DP ensures they cannot infer sensitive details like health status.
- Post-Processing Immunity: Applying any function to the output of a DP mechanism does not weaken privacy guarantees.
  Example: If a DP algorithm releases a noisy count, further statistical operations on this count do not leak additional private information.

Real-World Applications of Differential Privacy

Differential Privacy is used in various domains:

- **Tech Companies:** Google, Apple, Microsoft use DP to collect aggregate user data while preserving privacy.
- **Healthcare:** Hospitals and research institutions use DP to share medical statistics without compromising patient confidentiality.
- **Government:** The US Census Bureau implemented DP in the 2020 Census to protect respondent identities.
- **Finance:** Banks and fintech companies leverage DP to analyze transaction patterns without exposing individual users.
- **Machine Learning:** DP-SGD allows training models on sensitive data while maintaining privacy guarantees.

Implementing Differential Privacy in ML

- **TensorFlow Privacy (TF-Privacy):**
  - Extends TensorFlow with DP mechanisms.
  - Implements DP-SGD for training deep learning models.
  - Used in Google's research for privacy-preserving ML.
- **Opacus (PyTorch):**
  - Efficient implementation of DP-SGD in PyTorch.
  - Optimized for large-scale deep learning models.
  - Developed by Meta for privacy-aware ML applications.
- **IBM Diffprivlib:**
  - A library for DP in classical ML algorithms.
  - Provides DP-compliant versions of sklearn estimators.

Key Idea: These libraries integrate DP into ML workflows, allowing privacy-preserving training without modifying core architectures.